ELEC 451	Control Systems	3	CCEE 331
Year 3, Spring Semester (16 Credits)			
BIOM 418	Clinical Laboratory Instruments	3	BIOM 311
BIOM 440	Computational Neuroscience	3	BIOM 312 and MATH 311
CCEE 460	Artificial Intelligence	3	CCEE 214
CCEE 460L	Artificial Intelligence Lab	1	Co-req.: CCEE 460
ELEC 432	Power Electronics	3	ELEC 320
	Engineering Technical Elective 1	3	
Year 3, Summer Semester (3 Credits)			
BIOM 499	Co-op Work Experience	3	ENGL 217
The above 118 credits complete the requirements for a Bachelor of Science degree in Biomedical Engineering.			
Year 4, Fall Semester (16 Credits)			
BIOM 519	Advanced Therapeutic Devices	3	BIOM 414
BIOM 522	Advanced Medical Imaging	3	BIOM 421
BIOM 595A	BE Summative Learning Project 1	1	ENGL 217
CCEE 534	Signal Processing	3	CCEE 331
ENGR 510	Engineering Project Management	3	
	Engineering Technical Elective 2	3	
Year 4, Spring Semester (18 Credits)			
BIOM 531	Clinical Engineering	3	BIOM 417 BIOM 421
BIOM 595B	BE Summative Learning Project 2	3	BIOM 595A
	Engineering Technical Elective 3	3	
	Engineering Technical Elective 4	3	
	Engineering Technical Elective 5	3	
	Social Sciences/ Humanities Elective II	3	
Total BE credits: 152			

* BS-bound students are required to take BIOM 400 – Summative Learning (BS) Project instead.
Note: Engineering Technical Electives (levels 400 or above) are selected as such:
6 credits restricted Departmental Electives
9 credits from any Engineering discipline at the 400 level or above.

Courses Description

I. Mandatory Courses

Non-Major Courses

A description of the non-major mandatory courses follows.

BIOL 210	**Human Anatomy and Physiology and Lab**	**3(2,2)**
A general overview of cell structure and function, and the anatomy and physiology of human body systems. These include cardiovascular, central nervous, respiratory,		

urinary, digestive, immune, and musculoskeletal systems. The course also offers a set of experiments that deal with basic biological processes and advanced biological assessments. **Prerequisite**: None.

CCEE 214	Programming	3(2,2)
Understanding of the basic principles of programming and computer systems; Applications of programming to the solution of engineering problems; Control structures, functions, arrays, pointers, and structures.		

CCEE 221	Logic Design	3(3,0)
This course addresses Boolean algebra and logic simplification techniques, data representation, and the design of combinational logic networks for decoders, encoders, multiplexers, and demultiplexers. Design of sequential logic devices, including flip-flops, registers, and counters, as well as analysis of devices used to build logic networks, including programmable logic devices.		

CCEE 221L	Logic Design Lab	1(0,2)
This Lab covers design techniques and implementation of combinational and sequential logic circuits. Experiments include logic gates; the design and implementation of logic circuits; combinational logic circuits (decoders, encoders, multiplexers, demultiplexers, and adders); and the design of sequential logic devices using flip-flops, registers, and counters. **Prerequisite**: CCEE 221. **Prerequisite**: CCEE 221. Equivalent to COSC 351L.		

CCEE 331	Signals and Systems	3(3,0)
This course covers the mathematical description and classification of continuous and discrete signals and systems. Topics include signal types, Fourier series, Fourier transforms, Discrete-Time Fourier Transform (DTFT), Fast Fourier Transform (FFT), Laplace transforms, z-transforms, linear time-invariant systems, transfer functions, and state-space representation. **Prerequisite:** MATH 216.		

CCEE 426	Design of Embedded Systems	3(3,0)
This course addresses the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, operating systems and scheduling, system-level simulation, and interface-based design. **Prerequisite**: CCEE 221. Equivalent to COSC 356.		

CCEE 426L	Design of Embedded Systems Lab	1(0,2)
This lab complements topics covered in the CCEE 426 course. It involves the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Experiments include Design using finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, scheduling, system-level simulation, and interface-based design. **Co-requisite**: CCEE 426. Equivalent to COSC 356L.		

CCEE 460	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include intelligent agents, informed and uninformed search strategies, and adversarial search. **Prerequisite**: CCEE 214 or equivalent.		

CCEE 460L	Artificial Intelligence Lab	1(0,2)
This lab will help prepare students to handle the huge amount of data produced by both humans and machines today. The student will learn to use state-of-the-art AI tools to analyze data and make complex decisions based on that data using real-world practical examples and case studies. At the end of the lab, the student will be able to apply AI knowledge in their field at work or in day-to-day life to make better decisions. **Co-req.:** CCEE 460.		

CCEE 534	Signal Processing	3(3,0)
This course covers time series analysis, frequency analysis, time-frequency, and time-scale analysis. It also covers the design of digital filters and signal modeling. **Prerequisite**: CCEE 331.		

ELEC 200	Introduction to Electrical and Computer Engineering	3(3,0)
In this course, the student will be trained in core skills to excel in electrical, biomedical, computer, and communications engineering. The student will be trained in application programming using an appropriate programming environment. Students will build on the basics of Programming learned in the introductory programming course to implement Digital Signal Processing applications and perform data analysis. Students will also be introduced to basic lab equipment such as Oscilloscopes, Function Generators, and Multi-Meters. Students will be introduced to microcontroller-based projects to further explore the Electrical and Computer Engineering majors (CCE, BIOM, and ELEC). Throughout the course, students will work in multidisciplinary teams to study engineering cases for the community and implement a project to address a community need.		

ELEC 210	Electric Circuits	3(3,0)
This course covers circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, RL, and RLC circuits; transformers; sinusoidal (AC) steady-state analysis; power calculation; and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent.		

ELEC 320	Electronics	3(3,0)
A course on semiconductors, PN junctions, diode theory and circuits, bipolar junction transistors, transistor fundamentals, transistor biasing, amplifiers, JFETs, MOSFETs, and operational amplifiers. **Prerequisite**: ELEC 210		

ELEC 320L	Electronics Lab	1(0,2)
This is a Lab course with experiments in Electrical and Electronic Circuits. It covers passive electrical elements and sources; lab instruments; voltage divider circuits; Thevenin's theorem; RC circuits; diode circuits; Op-Amp circuits; BJT and MOSFET characteristics. **Co-requisite**: ELEC 320.		

ELEC 340	Engineering Electromagnetics	3(3,0)
Introduction to vector analysis; Electrostatic fields in vacuum and dielectrics; Conductors; Capacitance; Electrostatic energy and forces; Poisson's equation. Magnetic fields, Maxwell's equations, static electric and magnetic fields, boundary-value problems, Laplace's and Poisson's equations, and static electromagnetic fields. Time dependence in Maxwell's equations and plane-wave propagation. **Prerequisite**: PHYS 211.		

ELEC 451	Control Systems	3(3,0)
Introduction to control systems; Modeling of systems in various energy domains; transfer functions and block diagrams; time-domain analyses; Laplace-domain analyses; frequency-domain analyses; stability; precision; rapidity; root locus; Bode, Nyquist, and Nichols diagrams; design of PID controllers and dynamic compensators. **Prerequisite**: CCEE 331.		

ENGL 210	English Composition and Rhetoric	3(3,0)
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, a research paper, and an oral presentation. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	Professional English Communication	3(3,0)
This required course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics; present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements. **Prerequisite**: None.		

ENGR 510	Engineering Project Management	3(3,0)
The course covers key components of engineering project management including projects election and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics and professional conduct, realizing impact of engineering solutions in various contexts (global, economic, environmental, societal, etc...), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.		

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

PHYS 211	Physics: Electricity and Magnetism and Lab	3(3,1)
This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.		

PHYS 312	Modern Physics	3(3,0)
This course introduces the principles of revolutionary developments of the 20th century. It covers the interaction of light and matter (Photoelectric effect, Compton effect, Auger effect, etc.), the dual nature of light, various models of atomic description, quantum numbers, the relativistic approach, the Heisenberg Uncertainty Principle, the Schrödinger Equation, and an introduction to the band theory of solids and particle physics.		

Major Courses

Mandatory biomedical engineering courses are described below.

BIOM 301	Biomedical Sensors	3(3,0)
This course provides an introductory overview of the multidisciplinary field of sensors. The student is first introduced to the fundamental concepts of signals and noise in measurement, the characteristics of the measurement system, and the determination of the absolute quantity. Then the student learns about the different types of sensors to measure pressure, force, flow, motion, and temperature. **Prerequisite**: ELEC 210.		

BIOM 301L	Biomedical Sensors Lab	1(0,2)
The lab teaches students how to measure and interpret different types of signals. Experiments include measurements of signal, noise, pressure (strain gauge), and temperature; flow, motion, and force; and applications using research and simulation kits. **Co-requisite**: BIOM 301.		

BIOM 311	Biomedical Instrumentation	3(3,0)
The course describes the major medical electronic instruments used in diagnostic medicine. It explains the theoretical background, block diagrams, and circuitry of prosthetic devices, including different types of cardiac pacemakers, muscle and		

bladder prostheses, cochlear prostheses, and cardiac defibrillators and cardioverters. **Prerequisite**: BIOM 301.

BIOM 312	Biomedical Engineering Design	3(3,0)
This course examines the origins of various bioelectric phenomena and signals routinely recorded in modern clinical practice. Given adequate monitoring equipment, the engineer of today can record many forms of bioelectric phenomena with relative ease. This course describes the theoretical, block-diagram, and circuitry aspects of electrophysiological devices such as electrocardiogram, electroencephalogram, electroneurogram, electromyogram, and the like. **Prerequisite**: BIOM 301.		

BIOM 312L	Biomedical Engineering Design Lab	1(0,2)
The accompanying lab for BIOM 301 teaches students to measure and interpret biological and medical signals. Experiments cover the following topics: signal, noise, pressure (strain gauge), and temperature measurements; blood and gas flow, motion, and force measurements; and applications using electrodes based on medical equipment and research and simulation kits. **Co-requisite**: BIOM 312.		

BIOM 400	Summative Learning (BS) Project	3(3,0)
Team-oriented, project-based experience culminating in the creation of an artifact; milestones include project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisite**: ENGL 217 taken in the last semester of the BS Program. Annually.		

BIOM 313	Bio-electronics Circuit Design	3(3,0)
This course focuses on design and measurement processes for medical systems and associated constraints. Students are required to apply the learned concepts in projects to design healthcare technology. **Prerequisites**: BIOM 301, ELEC 320.		

BIOM 414	Advanced Biomedical Instrumentation	3(3,0)
The course describes the advanced medical electronic instruments used in therapeutic and diagnostic medicine. It includes extracorporeal shock wave lithotripsy, an infant incubator, a medical laser, a ventilator, and various medical devices used in the operating room, such as electrosurgical units and anesthesia units. **Prerequisite**: BIOM 311.		

BIOM 415L	Advanced Biomedical Lab	1(0,2)
The accompanying lab for BIOM 414 teaches students about advanced medical electronic instruments used in therapeutic and diagnostic medicine. Experiments cover the following topics: extracorporeal shock wave lithotripsy, infant incubator, medical laser applications, ventilator, and various medical devices used in the operating room, such as electrosurgical units and anesthesia units. **Co-requisite**: BIOM 414.		